

Fig. 3: Actual-size PCB of the timer circuit

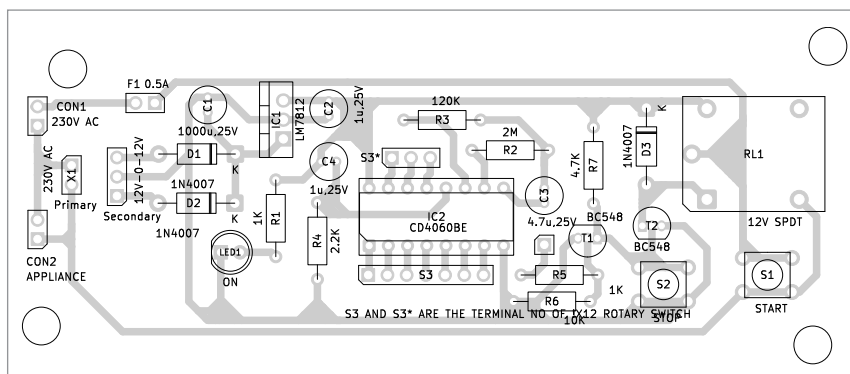


Fig. 4: Components side of the PCB

S2 can be pressed at any time to immediately turn off the appliance without waiting for the completion of the delay period.

Protective components such as flyback diode D3 connected across the relay coil ensure reliable operation by suppressing voltage spikes generated when the relay is switched off.

Construction and testing

An actual-size, single-sided PCB layout for the rotary switch-selectable timer circuit for industrial and domestic use is shown in Fig. 3, while the corresponding component layout is shown in Fig. 4.

After assembling the circuit on the PCB, mount it inside a suitable enclosure. Connect the primary and secondary terminals of the transformer to the PCB and fix the transformer at the bottom of the cabinet. Connect the fuse and appliance connectors on the front side of the cabinet. Mount the rotary switch S3 on the front panel and connect it to the PCB using wires with the respective contact numbers.

Care should be taken to keep the

oscillator components and IC connections short to avoid noise pickup and timing inaccuracies. The power supply and relay sections should be placed slightly away from the low-level timing circuitry to minimise electrical interference. Also, mount S1, S2, and LED1 on the front panel of the enclosure.

The rotary switch should be mounted firmly on the front panel and labelled clearly to indicate different timing selections. Start push-button S1 should be of a high-current-carrying type, while stop switch S2 can be a standard push-button. Proper insulation and spacing must be ensured in the mains and relay contact areas to maintain electrical safety, especially when controlling high-voltage loads.

During testing, the power supply section should be checked first to confirm correct voltage levels before inserting the IC. Once powered, the LED indicator should confirm normal operation. Each rotary switch position should be tested individually by pressing S1 and observing correct relay activation after the selected delay. S2 should be checked

ROTARY SWITCH S3 POSITION AND CD4060 OUTPUT DELAY

Rotary Position	CD4060 Output	Approx. Delay
1	Q4	~24 seconds
2	Q5	~48 seconds
3	Q6	~96 seconds (1.6 minutes)
4	Q7	~192 seconds (3.2 minutes)
5	Q8	~384 seconds (6.4 minutes)
6	Q9	~768 seconds (12.8 minutes)
7	Q10	~1536 seconds (25.6 minutes)
8	Q11	~3072 seconds (51.2 minutes)
9	Q12	~102 minutes (1.7 hours)
10	Q13	~204 minutes (3.4 hours)
11	Q14	~408 minutes (6.8 hours)

EFY Note: Changing the values of R2, R3, and C3 allows adjustment of the delay period.

PARTS LIST

Semiconductors:

- IC1 - LM7812, 12V voltage regulator
- IC2 - CD4060BE, 14-stage binary counter with oscillator
- T1, T2 - BC548 NPN transistor
- D1-D3 - 1N4007 silicon diode
- LED1 - 5mm red LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R5 - 1k Ω
- R2 - 2M Ω
- R3 - 120k Ω
- R4 - 2.2k Ω
- R6 - 10k Ω
- R7 - 4.7k Ω

Capacitors:

- C1 - 1000 μ F, 25V electrolytic
- C2, C4 - 1 μ F, 25V electrolytic
- C3 - 4.7 μ F, 25V electrolytic

Miscellaneous:

- RL1 - 12V SPDT relay
- S1 - Push-button switch 6A
- S2 - Push-button switch
- S3 - 12-position rotary switch
- X1 - 230V AC primary to 12-0-12V AC, 1A secondary step-down transformer
- F1 - 0.5A fuse
- CON1, CON2 - 2-pin connector
- AC appliance connector
- General-purpose PCB/ designed PCB/breadboard
- Enclosure, wires, and solder

for immediate reset and relay deactivation. After successful testing, the device is ready for use. **EFY**

Bonus. You can watch the video of the tutorial of this DIY project at <https://www.electronicsforu.com/videos-slideshows/make-your-own-rotary-switch-timer>

S.C. Dwivedi is an electronics enthusiast and circuit designer at EFY